PROJECT AND TEAM INFORMATION

Project Title

Smart Electricity Management System using Data Structures and Algorithms

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Electricity is used for many things at home. Appliances like AC, fan, heater, washing machine and computer can use different amounts of electricity.
Normally, we see the total amount on the electricity bill. We do not easily know how much electricity one particular appliance has used.
In our project, the user can enter the appliance name, start time and end time. The program will take the power and other details from its stored data.
It will then find the electricity used and the estimated cost. It will also check whether the appliance was used in peak or off-peak time.
The program can give a different time suggestion if using the appliance at that time is costly. This project also helps us apply C++ and DSA concepts to a practical problem.

State of the Art / Current Solution

Electricity usage can already be checked using normal meters, smart meters and electricity company applications. These systems mainly show the total usage or bill information.
For a normal user, it can still be difficult to find the usage of individual appliances. For example, the user may know the total bill but may not know how much of it came from an AC or heater.
Our system keeps appliance details in the program. It uses the selected appliance and its running time to calculate energy and cost.
It also checks peak and off-peak timings. Based on this, the system can show an alert and suggest a suitable time for using the appliance.

Project Goals and Milestones

The main aim of this project is to make a C++ based system for checking and managing electricity usage.
The project will include:

1. Store details of commonly used appliances.
2. Take appliance name, start time and end time from the user.
3. Find the energy used by the appliance.
4. Calculate the estimated electricity cost.
5. Check peak and off-peak usage.
6. Give an alert when peak time is used.
7. Suggest a better time when possible.
8. Give simple saving suggestions.
9. Show what-if results for different usage times.
10. Generate basic reports and budget alerts.

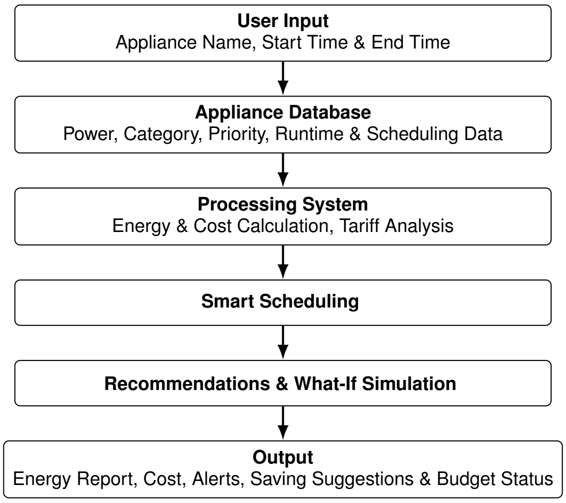
First, we will prepare the appliance data and basic requirements. Then we will create the C++ classes and data structures. After that, energy calculation, cost calculation and scheduling will be added. The last step will be testing the complete program and checking its output.

Project Approach

The project will be developed in C++. A predefined appliance database will be used. It will contain details like appliance ID, name, category, power, priority and usage information.
The user will select an appliance and enter its starting and ending time. The program will search the stored information and use the required values for calculation.
Classes and objects will be used to keep the program organised. Inheritance, abstraction and polymorphism may be used where they are useful.
For DSA, we will use STL structures such as vector, map, queue and priority queue. Searching and sorting will be used wherever required.
The program will compare the usage time with the stored tariff timings. It will calculate the cost and check whether the usage is during peak or off-peak hours.
File handling will be used for storing data and creating reports.

SYSTEM ARCHITECTURE AND OUTCOME

System Architecture (High Level Diagram)



System Flow:

Appliance Name, Start Time, End Time → Smart Scheduling → Appliance Database → Processing System → Energy & Cost Calculation → Tariff Analysis → Recommendations → What-If Simulation → Energy Report, Cost, Alerts, Saving Suggestions & Budget Status.

Project Outcome / Deliverables

The final project will be a working C++ program for electricity management.
It will contain a database of appliances and their basic details. The user can select an appliance and enter its running time.
The program will calculate energy consumption and cost. It will also show peak or off-peak usage and give an alert when needed.
The system will provide simple suggestions for reducing electricity use. It will also allow the user to compare different usage times through the what-if option.
The final work will include the source code, test cases, sample outputs, system design and project documentation.

Assumptions

The project will use predefined appliance and tariff information. There will be no direct connection with a smart meter or sensor.
The user will enter the appliance name, start time and end time manually. The power value will be taken from the stored appliance data.
The electricity cost will be calculated using the tariff values available in the program. Therefore, the result will depend on the data stored in the system.

References

1. Uttarakhand Electricity Regulatory Commission (UERC), *Tariff Order of UPCL for FY 2026–27*.
2. Uttarakhand Electricity Regulatory Commission (UERC), *Tariff and Regulatory Documents*.
3. C++ Standard Library documentation for STL containers, algorithms, file handling and exceptions.
4. Standard study material on Data Structures and Algorithms using C++.